

HyperGate: Hierarchical Perceptive Gating Network for Multi-domain Multi-task Recommendation

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Abstract

The growing prevalence of interactive pages in online services presents novel challenges for industrial recommendation systems, particularly in the realms of multi-domain and multi-task recommendations. These systems must not only model commonality across various domains but also account for distinctions between domains and tasks. However, the lack of explicit semantic information regarding domains and tasks complicates the process.

In this paper, we propose a novel parameter-sharing model for multi-domain multi-task recommendations, termed the **Hierarchical Perceptive Gating Network** (pronounced as **HyperGate**). Drawing inspiration from Collective Behavior Theory, our approach first augments semantic information for domains and tasks using contrastive learning, which then serves as perceptive control for the subsequent hierarchical gating network. The hierarchical gating network consists of three key components, including a domain-perceptive embedding gate, a domain-perceptive biasing gate, and a dual-perceptive fusion gate, thereby constructing a domain- and task-perceptive parameter-sharing network from the bottom up. Experiments conducted on two real-world datasets validate the

effectiveness of HyperGate, demonstrating substantial improvements in recommendation accuracy. Additionally, online A/B tests on industrial platform further validate the efficacy of HyperGate.

CCS Concepts

• **Information systems** → **Recommender systems**.

Keywords

Multi-domain Learning, Multi-task Learning, Recommendation System, Contrastive Learning, Feature Enhancement

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1 Introduction

With the rapid growth of web services and the explosion of information, recommendation systems have garnered significant attention as tools to understand user behavior and provide precise recommendations. Traditional recommendation systems are typically developed independently for each prediction task within each domain, where a single model is usually trained to fit the distribution of a single task in a domain [3, 12, 24, 25]. However, real-world platforms often encompass multiple domains. For instance, Huawei App Gallery platform includes domains such as *popular apps*, *premium apps*, and *daily picks*, as illustrated in Figure 1. Furthermore, each domain typically features various buttons for user interaction, leading to multiple prediction tasks aimed at modeling users' diverse interaction intentions. For example, Huawei App Gallery platform records user behaviors such as *download* and *register*.

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Figure 1: Example of the multiple domains and multiple tasks in some App platform. There are multiple app lists and types of interaction on Huawei App Gallery.

Traditional methods [3, 12, 24, 25, 33] separate the multi-domain multi-task recommendation (MDMTR) problem as multiple individual problems, optimizing each task in each domain independently. However, this approach has two main limitations. First, maintaining a set of model parameters for each domain and each task, which includes a large embedding table, increases the system’s complexity and reduces scalability. Second, there is an overlap of users and items across different domains and tasks, and shared commonalities can lead to improved performance. Moreover, training individual models with only the corresponding data may decrease accuracy while training collectively can get a more stable model.

To overcome these limitations, recent works have focused on designing model structures from the perspective of parameter sharing [27]. This approach not only reduces the number of parameters in online service models, but also fully exploits the inherent relationships between different scenarios and tasks. However, the variability in user behavior and item distribution across different domains and tasks results in a seesaw effect [19, 20], where parameter sharing fails to deliver consistent improvements across all domains and tasks simultaneously. Consequently, many approaches employ gating mechanisms [1, 13, 17, 34] to regulate the flow of information within the model in a domain- or task-specific manner, thus ensuring that the information is differentiated in various domains or tasks. For instance, in MMoe [17], a gate is used for each task to weight the output of shared experts; in DTRN [13], a gate is employed to refine underlying features specific to tasks; in PEPNet [1], a gate is used to reweight features relevant to domains. In summary, the gate mechanism is employed at the feature level to ensure domain relevance or task relevance, enabling the representations obtained under the parameter-sharing structure to possess both commonalities and distinctions.

Despite the advancements in leveraging gate mechanisms, these methods still exhibit significant limitations. First, these methods often overlook the challenge posed by the absence of explicit semantic information for domains and tasks, providing only basic indicators. Due to the limited information contained in indicators, the model’s ability to distinguish between domains and tasks will be limited, thereby reducing feature differentiation, and ultimately, a decline in overall model performance due to the seesaw issue. Secondly, existing models merely focus on differentiated representation extraction from the perspective of domain or task while lack

a dedicated structure to effectively discriminate between different joint distributions of tasks and domains, hindering their ability to capture the unique characteristics of various task-domain combinations. For example, models such as PEPNet and DTRN solely focus on domain-specific and task-specific regulation within the underlying embeddings, respectively, which hinders their ability to achieve enhancements across all domains and tasks simultaneously.

To address the aforementioned limitations, we propose a novel MDMTR framework called the Hierarchical Perceptive Gating Network (HiPerGate, pronounced as **HyperGate**). Inspired by Collective Behavior Theory [21], we introduce two augmentation modules to mitigate the absence of explicit semantic information for domains and tasks. These modules extract informative features from domain- and task-specific behaviors, serving as control signals for the hierarchical gating network. Contrastive learning [4, 18] is employed to optimize the augmentation process by aligning representations of behaviors within the same domain or task. The hierarchical gating network comprises three core components: domain-perceptive embeddings, domain-perceptive multi-experts, and a dual-perceptive fusion gate. The domain-perceptive embeddings and multi-experts utilize the gating mechanism to enhance the embedding table and multi-expert layer, leveraging augmented domain features to enable domain-specific feature interactions. The dual-perceptive fusion gate integrates representations from multiple experts with augmented domain and task features, enhancing the alignment between these features and the joint distribution of domain and task. This overall design empowers the network to discern subtle differences across diverse joint distributions of domain and task, further alleviating the seesaw issue. The contributions are summarized:

- We propose HyperGate, a novel framework for multi-domain multi-task recommendation. To address the lack of informative domain and task representations, we employ two augmentation modules to extract semantic representations. Subsequently, a hierarchical gating network is designed to capture nuanced distinctions across diverse domain-task combinations, thereby mitigating the seesaw effect.
- We conduct extensive experiments on two real-world industrial datasets, demonstrating that our HyperGate framework significantly outperforms state-of-the-art methods. The results validate the effectiveness of the proposed components.
- We successfully deploy HyperGate in the real-world industrial platform Huawei App Gallery, serving millions of active users. This deployment results in an average of 3.69% increase in effective Cost Per Mille (eCPM) within three representative domains.

2 Methodology

In this section, we describe our HyperGate structure in details. Following recent advanced works [10, 20, 26, 32, 34], HyperGate is built upon the MMoe [17] architecture due to its demonstrated effectiveness in addressing multi-task problems, despite its relatively long history. The overall framework is illustrated in Figure 2.

2.1 Problem Definition

Here we define the notations and problem settings of the multi-domain multi-task recommendations in our work. Let $\mathcal{U}$ and $\mathcal{I}$ denote the user set and item set. The MDMTR problem aims to

build and optimize a unified model to give predictions on T tasks on D domains simultaneously. All the users, items and tasks may overlap across different domains, and each sample is one interaction between a user and an item on a specific domain (e.g. a user downloads an app on the *premium app* list). Given a sample in domain $d \in \{1, \dots, D\}$, the input feature of the sample is defined as $\mathbf{x}_d = [\mathbf{x}_d^{\mathcal{U}}; \mathbf{x}_d^{\mathcal{I}}; \mathbf{x}_d^{\mathcal{C}}; \mathbf{x}_d^{\mathcal{D}}]$, consisting of user features $\mathbf{x}_d^{\mathcal{U}}$ (i.e. user id, age), item features $\mathbf{x}_d^{\mathcal{I}}$ (i.e. item id, category), context features $\mathbf{x}_d^{\mathcal{C}}$ (i.e. interaction timestamp), and domain-specific features $\mathbf{x}_d^{\mathcal{D}}$. The label of each sample consists of labels of all tasks $\mathbf{y}_x = (y_{x,1}, y_{x,2}, \dots, y_{x,T})$, where each label $y_{x,t}$ is a binary signal in $\{0, 1\}$ that denotes whether the user gives positive feedback or not in the task t . Our research goal is to learn a joint prediction function $\hat{\mathbf{y}}_x = (\hat{y}_{x,1}, \dots, \hat{y}_{x,T}) = f(\mathbf{x}_d)$ for each task t on each domain d .

2.2 Domain-Perceptive Embedding

In the context of MDMTR, the embedding table is shared among all domains and tasks while the significance of features can vary significantly across different domains. For example, when evaluating the same restaurant on a platform, the price might be a key factor on a budget-friendly list, whereas the rating could be more crucial for a list of top-rated restaurants. However, most methods fail to consider the various significances of the underlying embedding vector. Denoting $\mathbf{e}_x = E(\mathbf{x}) = E([\mathbf{x}^{\mathcal{U}}; \mathbf{x}^{\mathcal{I}}; \mathbf{x}^{\mathcal{C}}])$ as the embedding function in the following parts for convenience, we leverage the gating mechanism to implement a domain-perceptive adaptive embedding, consisting of a contrastive domain augmentation module and a domain-perceptive embedding gate.

2.2.1 Contrastive Domain Augmentation Module. To effectively capture the distinctions among various domains, an informative semantic domain representation is desirable. However, real-world applications only have limited domain features, mostly the indicator lacking rich semantic information. For instance, the App platform identifies domains primarily by ID without accompanying descriptive attributes.

Firstly, inspired by the Collective Behavior Theory [21], which posits that group characteristics can be inferred from member behaviors, we extract augmented domain representation from user behaviors through the following formulation:

$$\mathbf{z}_x = \mathbf{W}_2^{DA} \text{Sigmoid}(\mathbf{W}_1^{DA} \text{SG}(E(\mathbf{x})) + \mathbf{b}_1^{DA}) + \mathbf{b}_2^{DA} \quad (1)$$

where $\mathbf{W}_1^{DA} \in \mathbb{R}^{\tilde{m} \times m}$ and $\mathbf{W}_2^{DA} \in \mathbb{R}^{\tilde{m} \times \tilde{m}}$ are learnable weights, $\mathbf{b}_1^{DA}, \mathbf{b}_2^{DA} \in \mathbb{R}^{\tilde{m}}$ are learnable biases. $\text{SG}(\cdot)$ denotes the stop-gradient operation to isolate the augmentation module from influencing the underlying embedding.

To optimize the augmentation module, we pull samples within the same domain closer while separating those from different domains. We initialize an anchor embedding for each domain, denoted as $\tilde{\mathbf{E}}^{\mathcal{D}} \in \mathbb{R}^{D \times \tilde{m}}$, which can be interpreted as a global representation of the domain. For a sample $\mathbf{x}_d$ in domain d , its corresponding anchor is $\tilde{\mathbf{e}}_d = \tilde{\mathbf{E}}^{\mathcal{D}}(d)$, serving as a global domain representation. Inspired by the success of contrastive learning in recommendation [9, 15, 28], we treat the augmented representation $\mathbf{z}_x$ and the anchor representation $\tilde{\mathbf{e}}_d$ as two views of the same domain. To optimize the augmentation module, we employ the InfoNCE [18]

loss function to draw views of the same domain closer and repel those of different domains, as follows:

$$\ell_{DA} = -\frac{1}{|B|} \sum_{\mathbf{x}_d \in B} \log \frac{\exp \text{sim}(\mathbf{z}_{\mathbf{x}_d}, \tilde{\mathbf{e}}_d^{\mathcal{D}})}{\sum_{j \in \{1, \dots, D\}} \exp \text{sim}(\mathbf{z}_{\mathbf{x}_d}, \tilde{\mathbf{e}}_j^{\mathcal{D}})} \quad (2)$$

where $\tilde{\mathbf{e}}_j$ represents the anchor representation of domain j , B denotes the set of samples in a batch. $\text{sim}(\cdot)$ represents the similarity function, which is set as the inner product in our implementation.

2.2.2 Domain-Perceptive Embedding Gate. Given the shared embedding table across domains, we incorporate a gating mechanism to adaptively refine underlying embedding vectors to capture distinctions. We use the augmented domain feature $\mathbf{z}_x$ as the control signal of the gate, refining the embedding vector in an element-level manner. Specifically, we leverage a SENet-like [5] gate structure that has been widely applied in recommendation in recent years [1, 7, 13]. The gating mechanism comprises two linear layers sequentially equipped with ReLU and Sigmoid activations, respectively. The initial layer, acting as a squeeze layer, compresses features to capture global information. Subsequently, the excitation layer generates bit-wise importance weights through the Sigmoid activation. Given the input embedding vector $\mathbf{e}_x \in \mathbb{R}^m$ and the concatenated vector of augmented representation and the anchor representation $\mathbf{z}'_x = [\mathbf{z}_x; \tilde{\mathbf{e}}_d] \in \mathbb{R}^{2\tilde{m}}$, the gate structure is formalized as follows:

$$\mathbf{e}'_x = \gamma * \text{Sigmoid} \left(\mathbf{W}_2^{EG} \text{Relu} \left(\mathbf{W}_1^{EG} \mathbf{z}'_x + \mathbf{b}_1^{EG} \right) + \mathbf{b}_2^{EG} \right) \odot \mathbf{e}_x \quad (3)$$

where $\mathbf{W}_1^{EG} \in \mathbb{R}^{n \times 2\tilde{m}}$ and $\mathbf{W}_2^{EG} \in \mathbb{R}^{m \times n}$ are learnable weights, $\mathbf{b}_1^{EG} \in \mathbb{R}^n, \mathbf{b}_2^{EG} \in \mathbb{R}^m$ are learnable bias. The dimension n represents the compressed dimension, which is smaller than $\tilde{m}$ and m . The γ is a constant scaling factor. The output $\mathbf{e}'_x$ of the embedding gate is then leveraged as the domain-perceptive embedding.

2.3 Domain-Perceptive Multiple Experts

Considering the ability to capture the heterogeneity of various tasks in the multiple-experts structure in MMoE [17], we adopt the multiple-experts structure as well. However, MMoE exclusively focuses on multi-task scenarios and falls short of capturing the nuanced distinctions inherent in the various joint distributions of domains and tasks. Thus, we design domain-perceptive multiple experts regulated by biasing gates.

Concretely, each expert is implemented as a multi-layer perceptron (MLP), while biases vary across domains. We introduce a domain-perceptive biasing gate to generate expert biases based on the augmented domain representation. The output of the l -th layer of the k -th expert is calculated as follows:

$$\mathbf{h}_k^{(l+1)} = \sigma \left(\mathbf{W}_k^{(l+1)} \mathbf{h}_k^{(l)} + \mathbf{b}_k^{(l+1)} + \mathbf{b}_{\mathbf{x}_d}^{(l+1)} \right) \quad (4)$$

where the $\sigma(\cdot)$ is the non-linear function. $\mathbf{W}_k^{(l+1)}$ and $\mathbf{b}_k^{(l+1)}$ denote the learnable weights and bias, respectively. $\mathbf{b}_{\mathbf{x}_d}^{(l+1)}$ is the domain-perceptive bias generated by the biasing gate with a 2-layer MLP as follows:

$$\mathbf{b}_{\mathbf{x}_d}^{(l)} = \mathbf{W}_2^{BG, (l)} \sigma \left(\mathbf{W}_1^{BG, (l)} [\mathbf{z}_x; \tilde{\mathbf{e}}_d] + \mathbf{b}_2^{BG, (l)} \right) + \mathbf{b}_2^{BG, (l)} \quad (5)$$

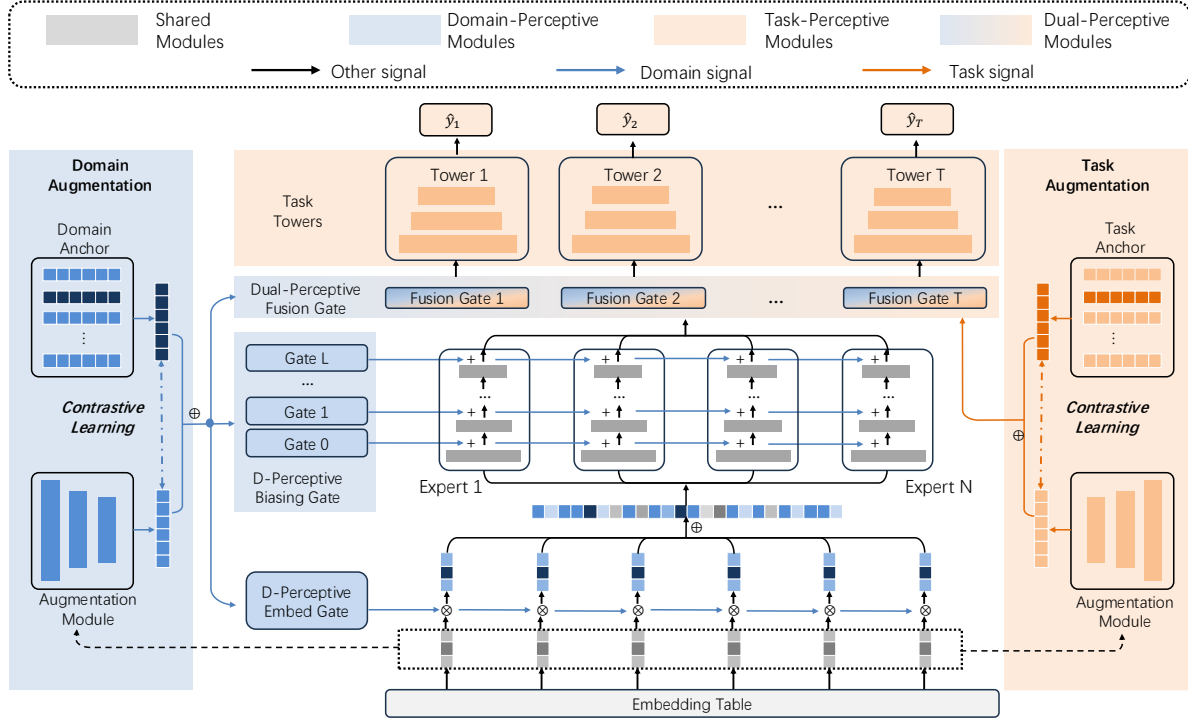


Figure 2: The overall framework of HyperGate. The blue and orange blocks represent the domain-perceptive modules and task-perceptive modules, respectively. The dashed line denotes stop-gradient forward.

2.4 Dual-Perceptive Fusion Gate

After obtaining outputs from each domain-perceptive expert, they are fused to form the input for task towers. While MMoE [17] introduced an enhanced multiple-gate design over the single-gate approach in MOE, the MDMTR setting requires considering the joint influence of domain and task during fusion. This section details an improved dual-perceptive fusion gate design.

2.4.1 Contrastive Task Augmentation Module. Similar to the challenge identified in Section 2.2.1, informative task features are often absent in MDMTR settings. For example, the industrial app platform lacks semantic distinctions between tasks like "download" and "registration." To address this, we adapt the domain augmentation approach to create a task augmentation module. By analyzing user behaviors within each task, we generate informative task representations from in-task behaviors. For instance, the "paid" task is more closely associated with "young students" and "gaming apps" than with "70s generation" and "news apps."

Specifically, the task augmentation module is a two-layer MLP. Given the feature input $\mathbf{x}$, the augmented task representation is:

$$\mathbf{t}_x = \mathbf{W}_2^{TA} \text{Sigmoid}(\mathbf{W}_1^{TA} \text{SG}(E(\mathbf{x})) + \mathbf{b}_1^{TA}) + \mathbf{b}_2^{TA} \quad (6)$$

where $\mathbf{W}_1^{TA} \in \mathbb{R}^{\tilde{m} \times m}$ and $\mathbf{W}_2^{TA} \in \mathbb{R}^{\tilde{m} \times \tilde{m}}$ are learnable weights, $\mathbf{b}_1^{TA}, \mathbf{b}_2^{TA} \in \mathbb{R}^{\tilde{m}}$ are learnable bias. Different from the optimization in the domain augmentation module, where each sample belongs to a single domain, a sample can be associated with multiple tasks in the task augmentation module. Consequently, the optimization

process for the task augmentation module diverges slightly from that of the domain augmentation module as follows:

$$\ell_{TA} = -\frac{1}{T} \sum_{i \in \{1, \dots, T\}} \log \frac{\exp \text{sim}(\bar{\mathbf{t}}_i, \tilde{\mathbf{e}}_i^T)}{\sum_{j \in \{1, \dots, T\}} \exp \text{sim}(\bar{\mathbf{t}}_i, \tilde{\mathbf{e}}_j^T)} \quad (7)$$

where $\bar{\mathbf{t}}_i = \frac{1}{\sum_{x \in B} y_{x,i}} \sum_{x \in B} y_{x,i} \mathbf{t}_x$ denotes the average of augmented representations of samples in current batch for task i , $\tilde{\mathbf{e}}_j^T$ represents the anchor representation of task j . B denotes the set of samples in a batch.

2.4.2 Dual-Perceptive Fusion Gate. Leveraging the augmented representations of domains and tasks, the fusion of experts can be relevant to the domain and task jointly. Given the concatenated domain representation $\mathbf{z}'_x = [\mathbf{z}_x; \tilde{\mathbf{e}}_d]$, the concatenated vector of the augmented task representation and the anchor task representation $\mathbf{t}'_x = [\mathbf{t}_x; \tilde{\mathbf{e}}_t]$, the weights for task- t 's tower can be formulated as:

$$\mathbf{a}_t = \mathbf{W}_{t,2}^{FG} \left(\mathbf{W}_{t,1}^{FG} [\mathbf{z}'_x; \mathbf{t}'_x] + \mathbf{b}_{t,1}^{FG} \right) + \mathbf{b}_{t,2}^{FG} \quad (8)$$

$$\alpha_{tk} = \frac{\exp(a_{tk})}{\sum_j \exp(a_{tj})} \quad (9)$$

where $\mathbf{W}_{t,1}^{FG} \in \mathbb{R}^{n \times 4\tilde{m}}$ and $\mathbf{b}_{t,1}^{FG} \in \mathbb{R}^{\tilde{m}}$ are learnable weights and bias for the task t . The N denotes the number of experts. The output vectors of these experts are then weighted summed using the calculated weights:

$$\tilde{\mathbf{h}}_t = \sum_{k \in \{1, \dots, N\}} \alpha_{tk} \mathbf{h}_k^L \quad (10)$$

The hybrid representation is then fed into the prediction tower of task t to obtain the predicted value $\hat{y}_{x,t} \in (0, 1)$.

2.5 Training Objective

The overall training objective consists of three components: CTR loss and two contrastive losses for control signal extraction networks. The first is the Binary Cross-Entropy (BCE) loss commonly used in CTR (Click-Through Rate) tasks. The loss is calculated as follows:

$$\ell_{CTR} = \frac{1}{|B|} \frac{1}{T} \sum_{x \in B} \sum_{t \in \{1, \dots, T\}} y_{x,t} \log(\hat{y}_{x,t}) + (1 - y_{x,t}) \log(1 - \hat{y}_{x,t}) \quad (11)$$

where B denotes the set of samples in a batch.

The losses in Equation (2) and Equation (7) are used as auxiliary losses, serving as training objectives for the augmentation modules. The final loss function is formulated as follows:

$$\ell = \ell_{CTR} + \lambda_1 \ell_{DA} + \lambda_2 \ell_{TA} \quad (12)$$

where λ_1 and λ_2 are hyper-parameters.

3 Experiments

In this section, we conduct extensive experiments to investigate the effectiveness of the proposed HyperGate, aiming to answer the following research questions:

- **RQ1:** How does HyperGate compare to state-of-the-art recommendation models, particularly in the challenging multi-domain, multi-task setting?
- **RQ2:** How does each gating component influence the performance of HyperGate?
- **RQ3:** How does the augmented feature module influence the performance of the hierarchical gating network?
- **RQ4:** How does HyperGate perform in a real-world online platform environment?

3.1 Experimental Settings

3.1.1 Datasets and Metrics. We evaluate the efficacy of HyperGate on a public recommendation dataset **KuaiRand-Pure**¹ containing 3 domains and 2 tasks and one offline dataset containing 3 domains and 7 tasks collected from our industrial platform **Huawei App Gallery**. The details of those datasets are shown in Appendix A.1. We evaluated models with the commonly used AUC metric.

3.1.2 Baselines. To validate the performance comprehensively, we select representative baselines for comparison, including single-domain single-task (SDST) models, single-domain multi-task (SDMT) models, multi-domain single-task (MDST) models, and multi-domain multi-task (MDMT) models.

- **SDST Models:** We mix the data in all domains, then train each task separately, and evaluate the model in each domain to report the multi-domain multi-task results. **DeepFM** [3], **DCN** [24] and **DCNv2** [25] are implemented.
- **SDMT Models:** We mix data in all domains and then train the multi-task models. Evaluations are conducted in each domain separately. **SharedBottom**, **MMoE** [17], **CGC** [20], **PFE** [14] and **DTRN** [29] are implemented.

- **MDST Models:** We mix data in all domains and then train each task separately. Evaluations are conducted in each domain separately. **STAR** [19] and **HMoE** [10] are implemented.

- **MDMT Models:** We mix data in all domains and then train multi-task models. Evaluations are conducted in each domain separately. **M2M** [31], **HiNet** [34] and **PEPNet** [1] are compared.

Details of baselines can be found in the Appendix A.2.

3.1.3 Implementation Details. The embedding size is set to 10 and 30 for KuaiRand-Pure and Huawei App Gallery for all methods, respectively. The numbers of shared experts are set to 4 and 5 in KuaiRand-Pure and Huawei App Gallery, respectively. The hidden layers of each expert are set to [256, 256] and [1024, 1024], respectively. The fusion gates are one-layer MLP with hidden size of 128 and 256 in KuaiRand-Pure and Huawei App Gallery. The hidden layers of task towers are set to [128, 128] and [1024, 512, 256] in KuaiRand-Pure and Huawei App Gallery. The hidden sizes $\tilde{m}$ of augmentation modules are set to 64 and 1024 in KuaiRand-Pure and Huawei App Gallery, respectively. The hidden size n of the embedding gate is set as the quarter of the dimension of the control signal. The weights λ_1 and λ_2 are set as 0.1 and 1.0, respectively. Hyper-parameter analysis about the λ is conducted and reported in Section 3.4. We use Adam as the optimizer with a learning rate 10^{-3} . The batch sizes are set to 1024 and 10000 in KuaiRand-Pure and Huawei App Gallery, respectively. The weight decay is tuned in $\{10^{-4}, 10^{-5}, 10^{-6}, 0\}$. More settings about the baselines are detailed in Appendix A.3. The implementation is based on RecStudio [11], released in <https://github.com/XuHwang/HyperGate>.

3.2 Overall Performance (RQ 1)

We compare our HyperGate model against the four kinds of baseline methods described above. The results, presented in Table 1, lead to the following findings:

Finding 1: Most SDMT, MDST, and MDMT methods outperform SDST methods, which can be attributed to the advantage of parameter-sharing structures, where knowledge is effectively shared and transferred across domains or tasks. However, an exception is observed with MDST methods on the Huawei App Gallery dataset, where their performance is consistently lower than that of SDST methods across all tasks. This discrepancy may be due to the difficulty of knowledge transfer between domains in the Huawei App Gallery caused by a lack of informative domain features.

Finding 2: Multi-task methods exhibit significant superiority over single-task methods on two datasets. This advantage is primarily due to the strong correlations among tasks, particularly for click and subsequent conversion tasks [23, 30]. For instance, users who download and register an app are more likely to make a payment compared to those who do not register.

Finding 3: MDMT baselines demonstrate inferior performance compared to SDMT methods, particularly on the Huawei App Gallery dataset. Despite incorporating a multi-domain setting, MDMT methods exhibit limited domain dependency due to a lack of informative domain features, thereby resulting in suboptimal performance for these meticulously designed models compared to SDMT models such as MMoE and DTRN. This aligns with comparisons between MDST and SDST methods, underscoring the critical role of robust domain representations beyond simple domain identifiers.

¹<https://kuairand.com>

Table 1: Performance comparison of different methods in terms of AUC on two industrial datasets. The bold and underline values represent the best and the second-best results, respectively. It is worth noting that an AUC increase of 0.001 can be considered a significant improvement in CTR prediction.

Category	Method	Huawei App Gallery Domain A							KuaiRand Domain 1	
		Activate	Auth	Pay	Register	Return	KeyAct1	Ac2Ret	Click	Longview
SDST	DCN	0.8674	0.8706	0.8449	0.8700	0.8017	0.8391	0.7026	0.7682	0.7787
	DeepFM	0.8676	0.8818	0.8398	0.8753	0.8040	0.8438	0.7028	0.7819	0.7938
	DCNv2	0.8679	0.8815	0.8431	0.8753	0.8030	0.8439	0.7025	0.7891	0.8028
SDMT	SharedBottom	0.8712	0.9108	0.8578	0.8826	0.8112	0.8560	0.7118	0.7897	0.8051
	CGC	0.8735	0.8898	0.8473	0.8820	0.8089	0.8535	0.7072	0.7909	0.8063
	MMoE	0.8718	0.9120	0.8566	0.8838	0.8107	0.8536	0.7125	0.7911	0.8074
	PFE	0.8721	0.9040	0.8513	0.8830	0.8100	0.8637	0.7109	<u>0.7916</u>	0.8072
	DTRN	0.8725	0.9100	<u>0.8589</u>	<u>0.8843</u>	<u>0.8120</u>	0.8554	<u>0.7125</u>	0.7914	0.8074
MDST	STAR	0.8643	0.8395	0.8373	0.8681	0.7994	0.8359	0.6937	0.7817	0.7931
	HMoE	0.8639	0.8702	0.8282	0.8661	0.7975	0.8375	0.6927	0.7911	0.8028
MDMT	M2M	0.8699	0.9052	0.8543	0.8800	0.8091	0.8535	0.7088	0.7904	0.8075
	HiNet	0.8684	0.9095	0.8580	0.8803	0.8092	0.8535	0.7087	0.7890	0.8047
	PEPNet	0.8724	0.9096	0.8564	0.8820	0.8102	0.8566	0.7116	<u>0.7916</u>	0.8070
MDMT	Ours	<u>0.8734</u>	0.9154	0.8593	0.8858	0.8135	<u>0.8593</u>	0.7141	0.7923	0.8088
Category	Method	Huawei App Gallery Domain B							KuaiRand Domain 2	
		Activate	Auth	Pay	Register	Return	KeyAct1	Ac2Ret	Click	Longview
SDST	DCN	0.8478	0.8601	0.8051	0.8608	0.7870	0.8297	0.7036	0.6388	0.6530
	DeepFM	0.8487	0.8648	0.8023	0.8685	0.7907	0.8373	0.7037	0.6883	0.6605
	DCNv2	0.8496	0.8631	0.8067	0.8685	0.7906	0.8362	0.7043	0.7055	0.6995
SDMT	SharedBottom	0.8564	0.8944	0.8251	0.8788	0.8023	0.8537	0.7133	0.7062	0.7180
	CGC	0.8583	0.8712	0.8159	0.8756	0.7999	0.8465	0.7097	0.7045	0.7151
	MMoE	<u>0.8571</u>	<u>0.8949</u>	0.8262	<u>0.8791</u>	<u>0.8026</u>	0.8494	<u>0.7147</u>	0.7093	0.7151
	PFE	0.8565	0.8870	0.8211	0.8778	0.8011	0.8477	0.7124	0.7116	0.7234
	DTRN	0.8552	0.8904	0.8266	<u>0.8791</u>	0.8013	0.8534	0.7142	0.7112	<u>0.7244</u>
MDST	STAR	0.8449	0.8413	0.7947	0.8616	0.7875	0.8302	0.6956	<u>0.7149</u>	0.7161
	HMoE	0.8445	0.8592	0.7823	0.8595	0.7851	0.8319	0.6966	0.7043	0.6843
MDMT	M2M	0.8532	0.8878	<u>0.8280</u>	0.8752	0.8000	0.8512	0.7119	0.7072	0.7177
	HiNet	0.8499	0.8912	0.8286	0.8750	0.7985	0.8505	0.7111	0.7133	0.7231
	PEPNet	0.8550	0.8913	0.8248	0.8758	0.7998	<u>0.8565</u>	0.7143	0.7002	0.7063
MDMT	Ours	0.8567	0.8982	0.8268	0.8804	0.8042	0.8583	0.7159	0.7171	0.7279
Category	Method	Huawei App Gallery Domain C							KuaiRand Domain 3	
		Activate	Auth	Pay	Register	Return	KeyAct1	Ac2Ret	Click	Longview
SDST	DCN	0.8641	0.8495	0.9030	0.8941	0.8763	0.8604	0.7364	0.6894	0.6551
	DeepFM	0.8637	0.8659	0.9061	0.8980	0.8775	0.8662	0.7359	0.7074	0.7114
	DCNv2	0.8640	0.8858	0.9096	0.8979	0.8777	0.8685	0.7359	0.7226	0.7227
SDMT	SharedBottom	0.8667	0.9196	0.9257	0.9058	0.8841	0.8810	0.7460	0.7220	0.7245
	CGC	<u>0.8680</u>	0.8958	0.9142	0.9040	0.8815	0.8762	0.7405	0.7244	0.7296
	MMoE	0.8674	<u>0.9221</u>	<u>0.9261</u>	0.9060	0.8843	0.8785	<u>0.7466</u>	0.7260	0.7304
	PFE	0.8679	0.9105	0.9193	0.9054	0.8835	0.8790	<u>0.7437</u>	0.7259	0.7295
	DTRN	<u>0.8680</u>	0.9184	0.9251	<u>0.9062</u>	<u>0.8848</u>	0.8807	0.7464	0.7265	0.7305
MDST	STAR	0.8633	0.8120	0.9066	0.8949	0.8762	0.8613	0.7353	0.7217	0.7278
	HMoE	0.8558	0.8759	0.8937	0.8827	0.8690	0.8561	0.7255	0.7215	0.7182
MDMT	M2M	0.8370	0.8996	0.9103	0.8839	0.8561	0.8511	0.7299	0.7255	0.7308
	HiNet	0.8332	0.9060	0.9122	0.8847	0.8562	0.8520	0.7291	0.7241	0.7268
	PEPNet	0.8667	0.9163	0.9218	0.9043	0.8830	<u>0.8826</u>	0.7450	<u>0.7269</u>	<u>0.7334</u>
MDMT	Ours	0.8684	0.9229	0.9274	0.9069	0.8854	0.8838	0.7478	0.7304	0.7360

Finding 4: Our HyperGate model consistently outperforms all baseline methods across all tasks and domains, validating the framework's effectiveness. By incorporating augmentation modules, we extract

more informative semantic features that serve as control signals for the hierarchical gates, thereby enhancing the model's ability to discriminate between domains and tasks.

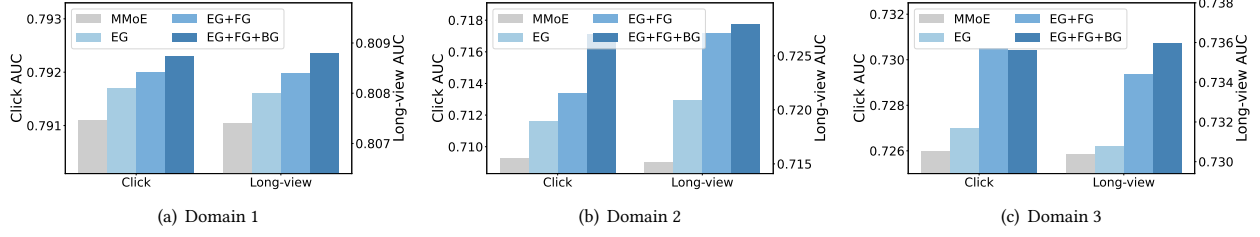


Figure 3: Ablation Study on KuaiRand-Pure. The base model is MMoE.

Table 2: Study on various control signals of the gates. T1 and T2 represent Click and Longview, respectively.

Type	Domain 1		Domain 2		Domain 3	
	T1	T2	T1	T2	T1	T2
base	0.7911	0.8074	0.7093	0.7227	0.7260	0.7304
x-gate	0.7909	0.8075	0.7125	0.7241	0.7244	0.7293
id-gate	0.7918	0.8079	0.7070	0.7189	0.7281	0.7312
augmented	0.7923	0.8088	0.7171	0.7279	0.7304	0.7360

3.3 Ablation Study (RQ 2 & RQ 3)

To isolate the contributions of HyperGate’s components, we conduct ablation studies on KuaiRand-Pure, examining the impact of gating and augmentation modules.

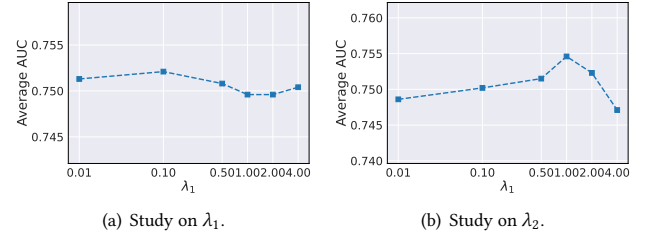
3.3.1 Study on Gating Modules (RQ 2). HyperGate comprises three gating modules: embedding gate (EG), biasing gate (BG), and fusion gate (FG). To evaluate their individual impact, we incrementally integrated each module into an MMoE baseline. Notably, FG replaces MMoE’s standard embedding with an augmented representation incorporating domain and task features. As Figure 3 demonstrates, performance consistently improves across all domains with each successive module addition. EG significantly enhances Domains 1 and 2 by refining domain-specific representations. The less pronounced improvement in Domain 3 may stem from the underfitting of augmented features due to a reduced number of behaviors compared to other domains. Incorporating FG and BG subsequently expands the model’s dependency on domain and task characteristics, thereby leading to increased performance.

3.3.2 Study on Augmentation Modules (RQ 3). Augmentation modules are pivotal components of HyperGate, as their outputs serve as control signals. To assess their impact, we conduct ablation studies by replacing augmented features with: (1) raw underlying embedding vectors e_x (*x-gate*); (2) learnable domain and task id embeddings (*id-gate*). Our proposed method is termed *augmented*. Table 2 presents results on KuaiRand-Pure. Despite leveraging our advanced hierarchical gate network, both x-gate and id-gate demonstrated only marginal improvements on specific tasks within certain domains, failing to achieve consistent performance enhancements across the board. This suggests that raw embeddings and simple ID representations lack the necessary discriminative information. In contrast, our augmentation module effectively aggregates domain

and task semantics from behavioral data, capturing granular distinctions. Consequently, the augmented model achieves superior performance in all tasks and domains.

3.4 Hyper-Parameter Analysis

To optimize the augmentation module, we introduce two additional contrastive loss terms and assign them with different weights λ_1, λ_2 . To further investigate the influence of these weights, we carry out experiments with various weights by selecting λ_1, λ_2 from $\{0.01, 0.1, 0.5, 1.0, 2.0, 4.0\}$ and calculate the average of AUC across all domains and tasks as the average AUC, visualized in Figure 4.

Figure 4: Hyper-parameter Analysis on λ_1 and λ_2

Due to the varying scales of ℓ_{DA} and ℓ_{TA} caused by different calculations, different λ values are required. A small λ disproportionately emphasizes CTR loss, resulting in underfitted augmentation modules and inadequate domain/task representations. Conversely, a large λ overweights the auxiliary loss, hindering backbone optimization and leading to a substantial AUC drop. To balance these factors, we empirically determined optimal peak values of ($\lambda_1 = 0.1, \lambda_2 = 1.0$).

3.5 Online A/B Testing (RQ 4)

To further validate the online performance, we deploy HyperGate in the advertising system on three representative app lists (domains) covering *activate*, *authorize*, *pay*, *register*, *return*, *keyac*, *ac2ret* tasks on Huawei App Gallery, and conduct a one-week online A/B test against baseline models, where millions of daily active users are involved. The results², presented in Table 3, shows significant comprehensive improvements across all domains in terms of effective Cost Per Mille (eCPM) and the click-through rate (CTR), further demonstrating the superiority of HyperGate.

²Due to system constraints, we are unable to isolate the individual gains in each task.

Table 3: Online gains in three representative domains.

Metrics	Domain A	Domain B	Domain C
eCPM	+3.98%	+4.88%	+2.22%
CTR	+0.24%	+2.23%	+4.32%

4 Related Works

4.1 Multi-domain Multi-task Recommendation

Multi-domain multi-task recommendation aims to model CTR prediction for multiple domains and tasks within a single model. It typically captures the commonalities and differences in behavior data across different domains or tasks through shared and domain-specific information. Some studies simplify the problem by considering it as either a multi-task or multi-domain issue.

Multi-task recommendation methods often overlook the domain information, focusing solely on various CTR or CVR tasks. These methods commonly employ parameter-sharing architectures to capture the commonalities among various tasks. The earliest proposed structure, SharedBottom, maintains a shared embedding table and underlying network for all tasks, while each task has an independent tower for final prediction. MoE improved upon SharedBottom by replacing the shared underlying network with multiple expert structures and using a gating mechanism to fuse the outputs of these experts. MMoE further enhanced the gating mechanism by maintaining a learnable gate structure for each task to obtain task-related fused representations. CGC [20] added task-specific experts to the MMoE framework, enabling better expression of both shared and task-specific representations. Building on these architectures, ESM [30] proposed a new optimization objective, replacing the original CVR objective with the optimization of CTCVR to leverage a large amount of CTR data for enhancing CVR model training. ESCM2 [23] addressed the estimation bias and independence priors in ESM using causal inference. Our work also adopts this parameter-sharing model architecture, and the proposed method is applicable to most of these frameworks.

Multi-domain recommendation, on the other hand, focuses on modeling a single task across different domains simultaneously, aiming to leverage more domain data to uncover commonality. Existing works mainly make improvements on multi-task models. For instance, HMoE [10] employs implicit MMoE and explicit MoE to extract commonalities and distinctions. STAR [19] adopts a topological network structure that splits network parameters into shared and domain-specific parts, merging these parameters into a new network during inference.

In recent years, some works have simultaneously considered both multi-domain and multi-task issues. M2M [31] proposed a meta-network to generate domain-specific parameters, explicitly handling features from different domains, and used an attention-based gate mechanism within the MMoE framework to address multi-task problems. HiNet [34] utilized a hierarchical MMoE structure to separately process domain and task information. PEPNet [1] used a gating mechanism at the lower levels to adjust features specific to domains and subsequently employed gates to obtain personalized network tower structures for multi-task prediction.

These methods all utilize domain information for multi-domain modeling; however, they fail to address the absence of explicit semantic information related to domains and tasks in the data, relying solely on ID information.

4.2 Gating Mechanisms in Recommendation

Gating mechanisms were originally inspired by biological neurology's gating mechanism, designed primarily to regulate the flow of information. They were first applied in natural language processing, with LSTM and GRU models being the most representative. Recently, many methods in recommendation systems have extensively applied gating mechanisms to regulate features. For instance, HGN [16] proposed gate structures at the feature and instance levels to adaptively select the importance of features and samples for sequential recommendation. FiBiNet [7] utilized a gating structure similar to SENet to adaptively adjust the weights of features, giving higher weights to more important features. GateNet [6] introduced feature embedding gates and hidden gates to adjust importance at the feature level and high-order feature interaction level, respectively. GDCN [22] extended the DCN [24] model using gating mechanisms to adaptively adjust the importance of feature interactions at different layers. Baidu's GemNN [2] inserted gate structures between the embedding layer and the MLP layer to filter out important feature information for subsequent network layers. In MMoE [17], a gating mechanism is used to allocate weights to representations extracted by different experts, ensuring task-specific representations. DTRN [17] employed gating mechanisms to adjust underlying representations, allowing each task to use distinct underlying representations. PEPNet [1] used gating mechanisms to adjust shared underlying representations into domain-specific representations based on domain information to model multi-domain information. However, these gating mechanisms often use the underlying features themselves as control signals, neglecting the importance of control signals. In multi-domain multi-task recommendations, they do not focus on more informative domain or task information to regulate the underlying representations.

5 Conclusion

This paper addresses the challenge of the seesaw phenomenon in multi-domain, multi-task recommendation, particularly in industrial settings where domain and task information is scarce. To overcome this, we propose HyperGate, a novel framework that employs two augmentation modules and a hierarchical gating network. The augmentation modules enrich semantic information by extracting domain and task representations from behavioral data. Subsequently, the hierarchical gating network effectively leverages these representations to enhance the model's ability to capture fine-grained distinctions between different joint distributions of domain and task. Extensive experiments on real-world industrial datasets validate HyperGate's effectiveness, and its practical impact is confirmed through online A/B testing on the real-world industrial platform Huawei App Gallery.

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Table 4: Dataset Statistics. NA indicates the absence of a user indicator within the dataset.

Dataset	KuaiRand-Pure			Industrial Dataset		
Domain	1	2	3	1	2	3
#Users	27,049	15,398	11,809	NA	NA	NA
#Items	7,580	6,233	4,633	695	702	1,088
#Instances	2,236,414	178,087	93,165	5,737,610	5,266,451	13,487,597
Percentage	89.18%	7.10%	3.72%	23.43%	21.50%	55.07%

A Experiments

A.1 Dataset Details

The statistics of those datasets are illustrated in Table 4.

- **KuaiRand-Pure**³: This dataset is collected from the short video platform Kuaishou⁴, consisting of 50 features. We use the “tab” as the domain indicator. The split ratio of training, validation, and test is 8:1:1. We use KuaiRand for short.
- **Industrial Dataset**: The dataset comprises one-month behavioral data collected from three lists within advertising system in the industrial platform, encompassing seven tasks: activate, authorize, pay, register, return, keyact, and ac2ret. List ID is employed as the domain indicator.

A.2 Baseline Description

The detailed description of all baselines are detailed here.

SDST Models: We mix the data in all domains and then train each task separately and evaluate the model in each domain to report the multi-domain multi-task results.

- **DeepFM** [3] is a pioneering method in recommendation, which combines the deep neural networks with Factorization Machine.
- **DCN** [24] replaces the Factorization Machine with a novel Cross Network, aiming to capture the linear cross features.
- **DCNv2** [25] is the state-of-the-art method in SDST models, which further improves the Cross Network in DCN.

SDMT Models: We mix data in all domains and then train the multi-task models. Evaluations are conducted in each domain separately.

- **SharedBottom** is a basic multi-task model that shares underlying embeddings among various tasks.
- **MMoE** [17] introduces a gate module for each task based on the MoE [8] structure.
- **CGC** [20] further improves MMoE by introducing additional task-specific experts to shared experts.
- **PFE** [14] combined task-specific experts and learned prototype features shared by all tasks.
- **DTRN** [29] propose a task-specific interest module and a task-specific representation refinement module to obtain task-specific bottom representation.

MDST Models: We mix data in all domains and then train each task separately. Evaluations are conducted in each domain separately.

- **STAR** [19] adopts a star-topology structure to train domain-specific and shared parameters.

- **HMoE** [10] is adapted from MMoE to share knowledge between various domains, then uses the hybrid of predictions of various domain towers as the final prediction.

MDMT Models: We mix data in all domains and then train multi-task models. Evaluations are conducted in each domain separately.

- **M2M** [31] leverages meta-units to generate domain-specific parameters with domain representation.
- **HiNet** [34] proposes a hierarchical network to extract domain-level and task-level information gradually.
- **PEPNet** [1] proposes EPNet and PPNet to obtain domain-relevant embeddings and refine the hidden representation of task towers with personalized information.

A.3 Implementation Details

We set the embedding dimension to 10 and 30 for all methods in KuaiRand and Huawei App Gallery, respectively. Regarding the task towers, we set the hidden layer as [256, 256, 256] and [1024, 512, 256] for all methods on KuaiRand and Huawei App Gallery. The number of cross layers for DCN and DCNv2 is set to 6 and 3 on KuaiRand and Huawei App Gallery, respectively. The hidden layers of shared layers in SharedBottom and experts in CGC, MMoE, HMoE, HiNet, M2M are set to [256, 256] and [1024, 1024] in KuaiRand and Huawei App Gallery, respectively. The number of experts (including shared experts and task-specific experts) is kept as 4 and 5 for each task in KuaiRand and Huawei App Gallery, respectively.

B Complexity Analysis

Since our HyperGate is built on MMoE, the additional gating modules will introduce more parameters and inference time. Therefore, we have analyzed the complexity of HyperGate from two perspectives, number of parameters and inference time, to demonstrate the efficiency of the proposed framework. First, we evaluated the inference time for HyperGate and MMoE in the same device on the Huawei App Gallery dataset, where HyperGate’s inference time is about 0.42ms per sample, compared to MMoE’s 0.37ms. For parameters excluding the underlying embedding table, HyperGate adds (a) an embedding gate, (b) a biasing gate, and (c) two augmentation modules beyond MMoE. With hidden size D , number of experts N , tasks T , and layers L , MMoE’s parameter is $(NL+TL+T) \cdot D^2 + TND$. And HyperGate adds (a) $0.5D^2 + (b)2LD^2 + (c)4D^2 = (4.5 + 2L)D^2$ extra parameters, which don’t scale with domains or tasks. For example, when $N = 5, T = 7, L = 3, D = 1024$, the number of HyperGate’s parameters is about 1.24 times that of MMoE. As the model becomes deeper and the task increases, i.e. L and T grow, the relative imp will further decrease.

³<https://kuairand.com>

⁴<https://kuaishou.com>